



Amrita School of AI
COIMBATORE CAMPUS

UNIT 2 · SESSIONS 4 AND 5

Vectors and matrices

Lists of complex numbers, and the machines that act on them.

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SESSION 4

Vectors in $\mathbb{C}^n$

Adding, scaling, combining, and turning a column on its side.

A quantum state is a column of complex numbers.

That column is a vector. Today you learn what you can do with one.

A column vector

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

Each entry v_k is a complex number.

The set of all such columns with n entries is called $\mathbb{C}^n$.

$$\begin{pmatrix} 1 \\ i \end{pmatrix} \in \mathbb{C}^2 \qquad \begin{pmatrix} 2 \\ -1 + i \\ 0 \end{pmatrix} \in \mathbb{C}^3$$

Everything happens entry by entry

Add two vectors of the same length:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$$

Scale by a complex number c :

$$c \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} c v_1 \\ c v_2 \end{pmatrix}$$

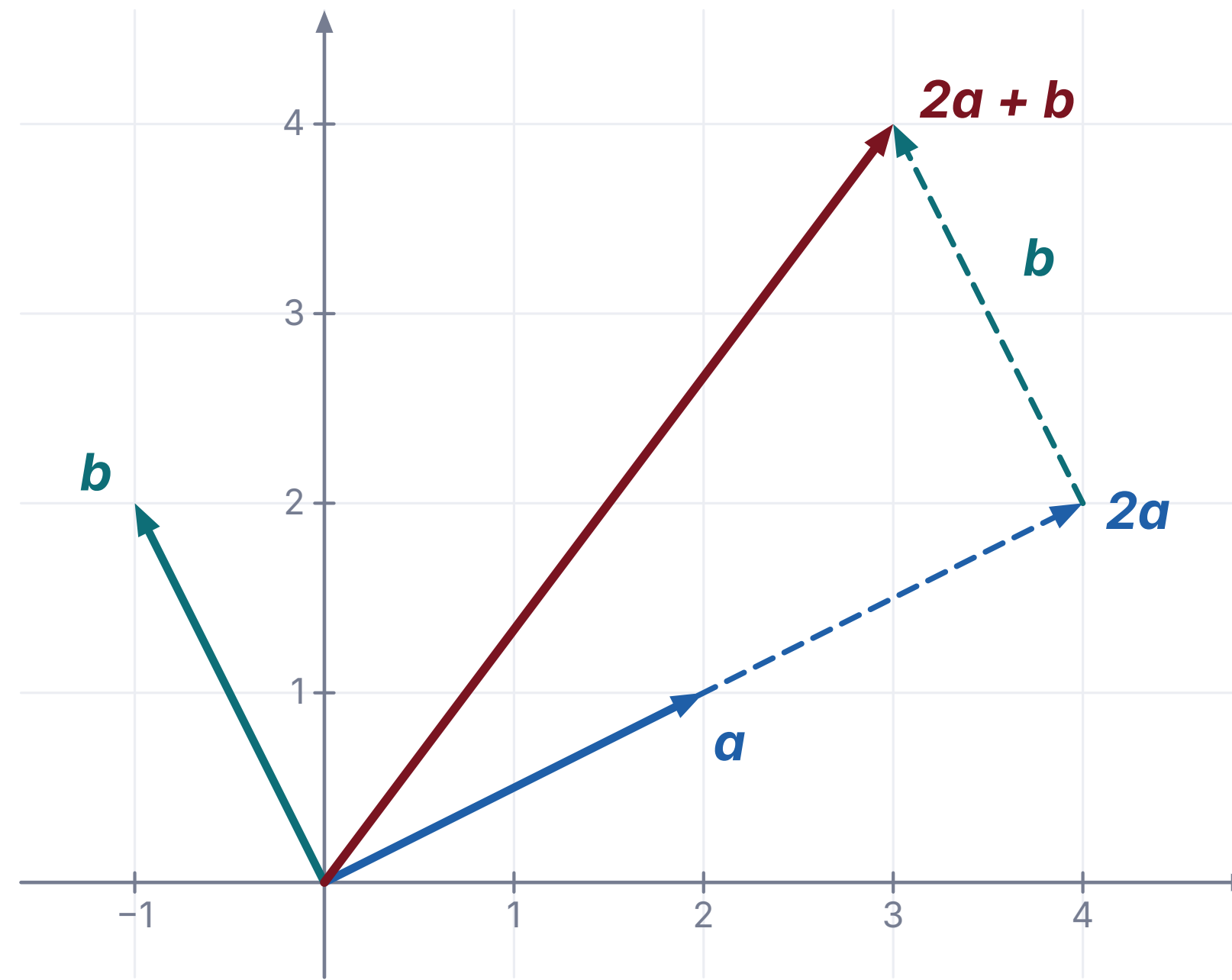
Scale $\begin{pmatrix} 1 \\ 2 - i \end{pmatrix}$ **by** i

Multiply **every** entry by i , using the arithmetic from Session 2.

The second entry: $i(2 - i) = 2i - i^2 = 1 + 2i$.

$$i \begin{pmatrix} 1 \\ 2 - i \end{pmatrix} = \begin{pmatrix} i \\ 1 + 2i \end{pmatrix}$$

Scale some vectors, then add them



The picture is real. The algebra works for complex vectors too.

A **linear combination** of $\vec{v}_1, \dots, \vec{v}_k$ is

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k$$

for any complex numbers $c_1, \dots, c_k$.

You cannot draw $\mathbb{C}^2$: it needs four real axes.

So we draw real vectors to build intuition, and trust the entry-by-entry algebra for complex ones.

SESSION 4 · WORKED EXAMPLE

$$\vec{u} = \begin{pmatrix} 1 \\ i \end{pmatrix}, \vec{v} = \begin{pmatrix} i \\ 1 \end{pmatrix}. \text{ Find } 2\vec{u} - i\vec{v}.$$

Scale each vector first:

$$2\vec{u} = \begin{pmatrix} 2 \\ 2i \end{pmatrix}, \quad i\vec{v} = \begin{pmatrix} i^2 \\ i \end{pmatrix} = \begin{pmatrix} -1 \\ i \end{pmatrix}$$

Then subtract entry by entry:

$$2\vec{u} - i\vec{v} = \begin{pmatrix} 2 - (-1) \\ 2i - i \end{pmatrix} = \begin{pmatrix} 3 \\ i \end{pmatrix}$$

Every vector in $\mathbb{C}^2$ is built from two

$$\begin{pmatrix} a \\ b \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Call these $\vec{e}_1$ and $\vec{e}_2$. The entries a and b are the amounts of each.

In Unit 3 the same two vectors are the qubit states $|0\rangle$ and $|1\rangle$.

Turn the column on its side

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$\vec{v}^T = (v_1 \quad v_2 \quad v_3)$$

The **transpose** is a row with the same entries.

Transpose, and conjugate every entry

$$\vec{v}^\dagger = (v_1^* \quad v_2^* \quad v_3^*)$$

Read it “v dagger”. This is the **conjugate transpose**.

Some texts write $\vec{v}^H$ or $\vec{v}^*$ for the same thing. This course uses $\dagger$ throughout, as quantum mechanics does.

$$\vec{v} = \begin{pmatrix} 1 + i \\ 2 \\ -3i \end{pmatrix}. \text{ Find } \vec{v}^T \text{ and } \vec{v}^\dagger.$$

Transpose: same entries, now a row.

$$\vec{v}^T = (1 + i \quad 2 \quad -3i)$$

Dagger: also flip the sign of every imaginary part.

$$\vec{v}^\dagger = (1 - i \quad 2 \quad 3i)$$

The dagger is easy to forget

- **Why it slips**

For **real** vectors $\vec{v}^T = \vec{v}^\dagger$, so the conjugate never mattered before.

With complex entries, a missed conjugate gives a plausible wrong answer.

- **The habit**

Dagger means **two** steps: transpose, then conjugate.

Doing it twice gets you back: $(\vec{v}^\dagger)^\dagger = \vec{v}$.

SESSION 4 · TRY THIS NOW

$$\vec{u} = \begin{pmatrix} 2 \\ i \end{pmatrix} \text{ and } \vec{v} = \begin{pmatrix} 1 - i \\ 3 \end{pmatrix}$$

Find:

- $\vec{u} + \vec{v}$
- $(1 + i) \vec{u}$
- $\vec{v}^\dagger$

Three minutes. Write each answer the right way up: a column or a row.

$$\vec{u} = \begin{pmatrix} 2 \\ i \end{pmatrix} \text{ and } \vec{v} = \begin{pmatrix} 1 - i \\ 3 \end{pmatrix}$$

$$\begin{aligned} \vec{u} + \vec{v} &= \begin{pmatrix} 3 - i \\ 3 + i \end{pmatrix} \\ \vec{v}^\dagger &= (1 + i \quad 3) \end{aligned}$$

$$(1 + i) \vec{u} = \begin{pmatrix} 2 + 2i \\ i + i^2 \end{pmatrix} = \begin{pmatrix} 2 + 2i \\ -1 + i \end{pmatrix}$$

A column is an array of shape (n, 1)

```
import numpy as np

u = np.array([[2], [1j]])          # column vector, shape (2, 1)
v = np.array([[1 - 1j], [3]])
print(u + v)                      # [[3.-1.j] [3.+1.j]]
print((1 + 1j) * u)               # [[ 2.+2.j] [-1.+1.j]]
print(v.conj().T)                 # [[1.+1.j 3.-0.j]] the dagger: a row
```

`.T` is the transpose. `.conj()` conjugates every entry. The dagger needs both.

Before Session 5

With $\vec{a} = \begin{pmatrix} 1 \\ -i \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 2i \\ 1 + i \end{pmatrix}$, find $3\vec{a} + \vec{b}$, $i\vec{a} - \vec{b}$ and $\vec{b}^\dagger$.

Check them in NumPy.

Part A of the Qubit Workbook is due at the **end of this week**.

Next: matrices, the objects that act on vectors.

SESSION 5

Matrices

A matrix takes a vector in and gives a vector out.

A grid of complex numbers

$$A = \begin{pmatrix} 1 & i & 0 \\ 2 & 3 & -i \end{pmatrix}$$

This one is 2×3 : two rows, three columns.

The entry in row j , column k is A_{jk} .

Here $A_{12} = i$ and $A_{23} = -i$.

A column vector is just an $n \times 1$ matrix.

Matrix times vector: the row view

Entry j of $A\vec{v}$ is **row** j of A multiplied into $\vec{v}$: pair up, multiply, add.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{pmatrix} = \begin{pmatrix} 17 \\ 39 \end{pmatrix}$$

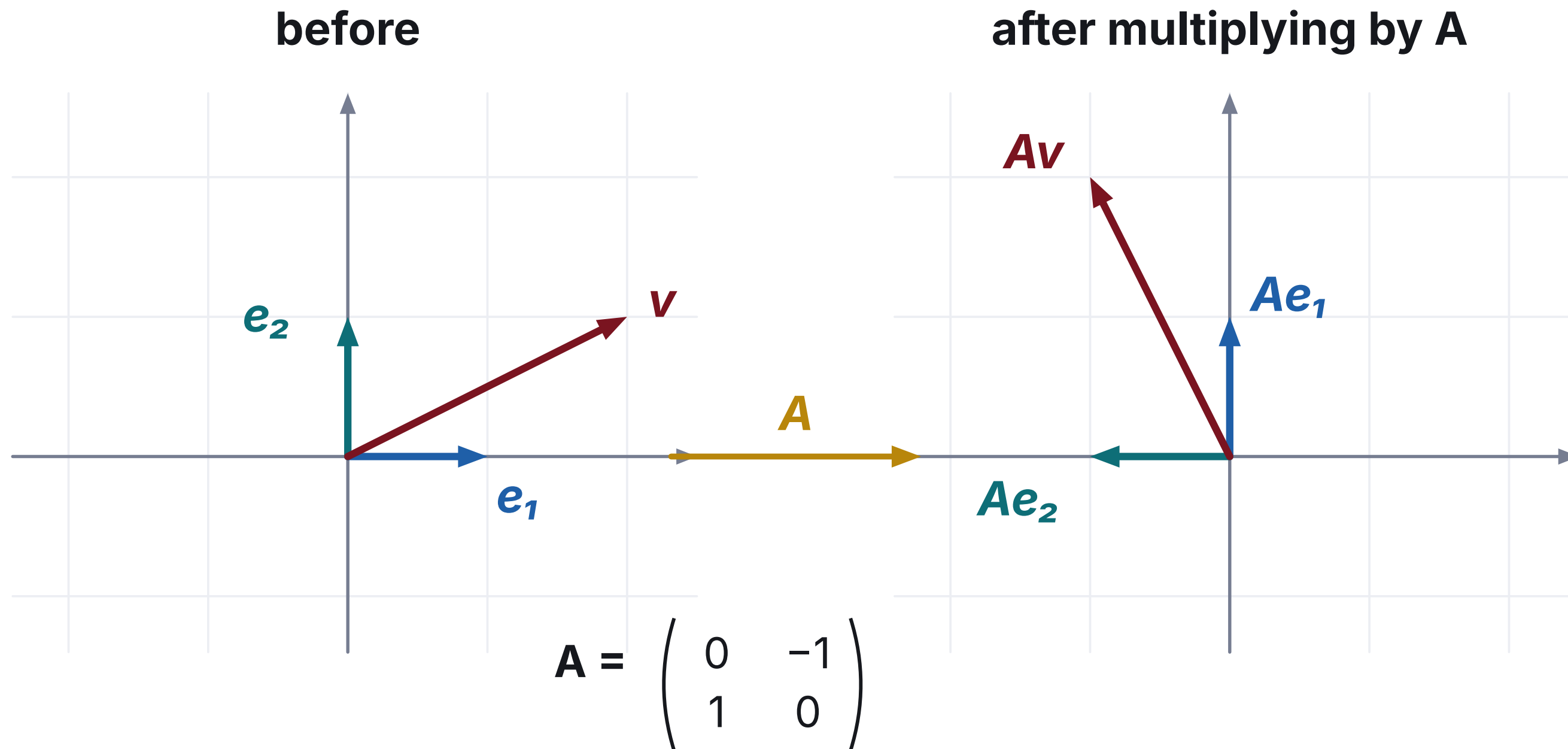
The same product: the column view

The product $A\vec{v}$ is a **linear combination of the columns** of A , with the entries of $\vec{v}$ as the amounts.

$$\begin{aligned} 5 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 6 \begin{pmatrix} 2 \\ 4 \end{pmatrix} \\ = \begin{pmatrix} 5 + 12 \\ 15 + 24 \end{pmatrix} = \begin{pmatrix} 17 \\ 39 \end{pmatrix} \end{aligned}$$

Same answer. Use whichever view helps.

A matrix moves every vector



The columns of A are where e_1 and e_2 land.

A complex matrix on a complex vector

$$\begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Row 1 with the vector, then row 2 with the vector.

$$= \begin{pmatrix} 1 \cdot 1 + i \cdot i \\ -i \cdot 1 + 2 \cdot i \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix}$$

Note $i \cdot i = -1$ cancels the 1.

The shapes must fit

- **Undefined**

A 2×3 matrix times a vector in $\mathbb{C}^2$.

Each row has three entries; the vector has two.

- **The rule**

$$(m \times n)(n \times 1) = (m \times 1)$$

Inner sizes must match. Outer sizes give the shape of the answer.

Every entry is a row times a column

$$\begin{array}{|c|c|} \hline A & \\ \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \times \begin{array}{|c|c|} \hline B & \\ \hline 0 & 1 \\ \hline 1 & 0 \\ \hline \end{array} = \begin{array}{|c|c|} \hline AB & \\ \hline 2 & 1 \\ \hline 4 & 3 \\ \hline \end{array}$$

row 2 of A
with column 1 of B:
 $3 \cdot 0 + 4 \cdot 1 = 4$

Entry (j, k) of AB = row j of A times column k of B

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \text{ Find } AB.$$

Column k of AB is A times column k of B .

So you can reuse matrix times vector twice.

$$\begin{aligned} AB &= \begin{pmatrix} 1 \cdot 0 + 2 \cdot 1 & 1 \cdot 1 + 2 \cdot 0 \\ 3 \cdot 0 + 4 \cdot 1 & 3 \cdot 1 + 4 \cdot 0 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} \end{aligned}$$

Now the other order: BA

$$BA = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$$

Multiplying by B on the right swapped the **columns** of A .

On the left, it swapped the **rows**.

Different matrices.

$$AB \neq BA$$

Order matters. In Unit 3, “do B , then A ” is the product AB , and swapping the order changes the result.

The matrix that does nothing

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Ones down the diagonal, zeros elsewhere.

$$I\vec{v} = \vec{v}$$

$$AI = IA = A$$

Write I_2, I_4 when the size matters.

The matrix that undoes A

$$A^{-1}A = AA^{-1} = I$$

For a 2×2 matrix there is a formula.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

It exists only when $ad - bc \neq 0$. That number is the **determinant**.

Invert $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$

Swap the diagonal, negate the other two, divide by the determinant:

$$ad - bc = 6 - 5 = 1$$
$$A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$$

Always check:

$$AA^{-1} = \begin{pmatrix} 6 - 5 & -2 + 2 \\ 15 - 15 & -5 + 6 \end{pmatrix} = I \checkmark$$

Invert $A = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$

The determinant:

$$ad - bc = 1 - i^2 = 2$$
$$A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$$

Check:

$$\frac{1}{2} \begin{pmatrix} 1 - i^2 & -i + i \\ i - i & -i^2 + 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I \checkmark$$

The dagger of a matrix

$$A = \begin{pmatrix} 1 & 2 + i \\ 3i & 4 \end{pmatrix}$$

Rows become columns: $(A^T)_{jk} = A_{kj}$. Then conjugate.

$$A^\dagger = \begin{pmatrix} 1 & -3i \\ 2 - i & 4 \end{pmatrix}$$

The $3i$ moved to the top right and changed sign.

The dagger of a product reverses the order

$$(AB)^\dagger = B^\dagger A^\dagger$$

The same holds for the transpose: $(AB)^T = B^T A^T$.

With $A = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & i \end{pmatrix}$:

$$(AB)^\dagger = B^\dagger A^\dagger = \begin{pmatrix} 1-i & 2 \\ -1 & -2i \end{pmatrix}$$

In the wrong order, $A^\dagger B^\dagger = \begin{pmatrix} 1 & 1 \\ -i & -3i \end{pmatrix}$.

SESSION 5 · TRY THIS NOW

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Find:

- AB and BA
- A^{-1} , and check it

Five minutes. Is $AB = BA$ here?

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

Not equal.

The determinant is $1 \cdot 1 - 1 \cdot 0 = 1$, so

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

Matrix product is @

```
import numpy as np

A = np.array([[1, 2], [3, 4]])
B = np.array([[0, 1], [1, 0]])
print(A @ B)           # [[2 1] [4 3]]
print(B @ A)           # [[3 4] [1 2]]
print(np.linalg.inv(np.array([[2, 1], [5, 3]]))) # [[ 3. -1.] [-5.  2.]]
M = np.array([[1, 2 + 1j], [3j, 4]])
print(M.conj().T)       # the dagger: [[1.-0.j 0.-3.j] [2.-1.j 4.-0.j]]
```

PITFALL

`A * B` multiplies entry by entry. It is **not** the matrix product.

Before Session 6

With $A = \begin{pmatrix} 2 & i \\ 0 & 1 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 1 \\ 1 + i \end{pmatrix}$:

find $A\vec{v}$, A^2 , A^{-1} and $A^\dagger$.

Check each one with `@`, `np.linalg.inv` and `.conj().T`.

Next: lengths and angles for complex vectors, the inner product.